Apache Spark:
The Ultimate Panacea for the Big Data Era

Apache Spark is a data analysis engine based on Hadoop MapReduce, which helps in the quick processing of Big Data. It overcomes the limitations of Hadoop and is emerging as the most popular framework for analysis.

With the advent of new technologies, the data generated by various sources such as social media, Web logs, IoT, etc, is proliferating in petabytes. Traditional algorithms and storage systems aren't sophisticated enough to cope with this enormous volume of data. Hence, there is a need to address this problem efficiently.

Introducing the Apache Spark engine

Apache Spark is a cluster computing framework built on top of Apache Hadoop. It extends the MapReduce model and allows quick processing of large volumes of data significantly faster, as data persists in-memory. It has fault tolerance, data parallelism capabilities and supports many libraries such as GraphX (for graph processing), MLlib (for machine learning), etc. These features have led to Spark emerging as the most popular platform for Big Data analytics and it being used by the chief players in the tech industry like eBay, Amazon and Yahoo.

Spark was created in 2009 by Matei Zaharia at UC Berkeley's AMPLab as a lightning fast cluster computing framework. In 2010, it was donated to the Apache Software Foundation under a BSD licence and has since been developed by contributors throughout the world. In November 2014, Zaharia's enterprise, Databricks, sorted a large dataset in record time by using the Spark engine. Spark 2.0.0 is the latest release, which came out on July 26, 2016.

Hadoop has been widely used due to its scalability, flexibility and the MapReduce model, but it is losing its popularity to Spark since the latter is 100x faster for in-memory computations and 10x faster for disk computations. Data is stored on disk in Hadoop but in Spark, it's stored in memory, which reduces the IO cost. Hadoop's MapReduce can only re-use the data by writing it to an external storage and fetching it when needed again. Iterative and interactive jobs need fast responses, but MapReduce isn't satisfactory due to its replication, disk IO and serialisation. Spark uses RDD (Resilient Distributed Dataset), which allows better fault tolerance than Hadoop, which uses replication. Though Spark is derived from Hadoop, it isn't a modified version of it. Hadoop is a method to implement Spark, which has its own cluster management system and can run in